F78PA Probability & Statistics A

Tutorial 7

1. Two random variables X and Y have joint probability mass function $p_{X,Y}(x,y)$ given by the following table

	$y=-1$	$y=0$	$y=1$
$x=-1$	0	0.25	0
$x=0$	0.25	0	0.25
$x=1$	0	0.25	0

   Determine the following:

   (a) Calculate $\mathbb{E}(X)$ and $\mathbb{E}(Y)$

   (b) Calculate $\mathbb{E}(XY)$

   (c) Are X and Y independent?

2. Evaluate the following integrals:

   (a)
   $$\int_0^1 \int_0^1 x^2 + y \, dxdy$$

   (b)
   $$\int_2^5 \int_1^3 xy \, dxdy$$

   (c)
   $$\int_0^1 \int_0^5 (x+y)^2 \, dydx$$

   (d)
   $$\int_1^2 \int_0^5 \frac{1}{(x+y)^3} \, dydx$$

   (e) Let $R = [0,3] \times [0,5]$
   $$\iint_R x \, dxdy$$

3. Evaluate the following integrals:

   (a)
   $$\int_0^1 \int_0^x 2x + 5y \, dydx$$

   (b) Let $R = \{(x,y) : 0 \leq x \leq y \leq 2\}$
   $$\iint_R e^{x+y} \, dxdy$$

4. Let $R = \{(x,y) : x^2 + 2y^2 \leq 1, x \geq 0\}$ and evaluate
   $$\iint_R x \, dxdy.$$

5. The Random variables X and Y have joint density function
   $$f_{X,Y}(x,y) = \frac{1}{y} e^{-y - \frac{x}{y}}, \quad 0 \leq x, y < \infty.$$

(a) Verify that $f_{X,Y}$ is indeed a joint density function.

(b) Calculate the probability that $X \leq Y$.

6. An insurance company models the joint distribution of the random variables X and Y, where:

   - X is the loss amount (in thousands of dollars) for a particular type of insurance policy and follows an exponential distribution with rate 2, i.e., $X \sim \text{Exp}(2)$.
   - Y is the deductible applied to the claim (in thousands of dollars) and follows a uniform distribution between 0 and b, i.e., $Y \sim \text{U}(0, b)$.

   The joint probability density function, $f_{X,Y}$, of X and Y is given by:

$$f_{X,Y}(x, y) = \begin{cases} \frac{2}{b}e^{-2x}, & x > 0 \text{ and } 0 \leq y \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

   The insurance company is interested in determining the probability that the deductible exceeds the loss by more than 1,000 dollars, i.e., $\mathbb{P}(Y - X > 1)$.

   Find this probability as a function of b by integrating the joint probability density function over the appropriate region.

Exam Style Question

Let X and Y be continuous random variables with joint probability density function:

$$f(x, y) = \begin{cases} 6(1 - y), & 0 \leq x \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Verify that $f(x, y)$ is a valid joint probability density function. [3 marks]

(b) Compute the probability that $X \leq \frac{1}{2}$ and $Y \geq \frac{1}{2}$. [4 marks]

(c) Find the probability that $X + Y \leq 1$. [3 marks]